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14. Java Exception Handling and Multi-Threading

14.1 Multithreading

Java provides built in support for multithreaded programming concept. A multithreaded program contains two or more parts that can run concurrently.
Each part of program is called a thread and each thread defines a seprate path of execution.
Thus, multithreading is form of multitasking.
It is also similar to dividing a task into subtasks and assigning them differently for execution independently and simultaneously.

There are two types of Multitasking:

1. Process based:

A process is a program that is executing. Thus, a process -based multitasking is the feature that allows the computer to run two or more programs concurrently.
In the process based multitasking, a program is a smallest unit of code that can be dispatched by the scheduler.

2. Thread based:

In a thread based multitasking environment, the thread is the smallest unit of dispatchable code.
That means a single program can perform two or more task

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simultaneously.

The threads are light weight processes. They share the same address space and cooperatively share the same heavy weight process

LifeCycle of a Thread :

The thread can enter into many states during its lifetime
The five types of states are:

i) Newborn state

ii) Runnable

iii) Running state

iv) Blocked state

v) Dead state

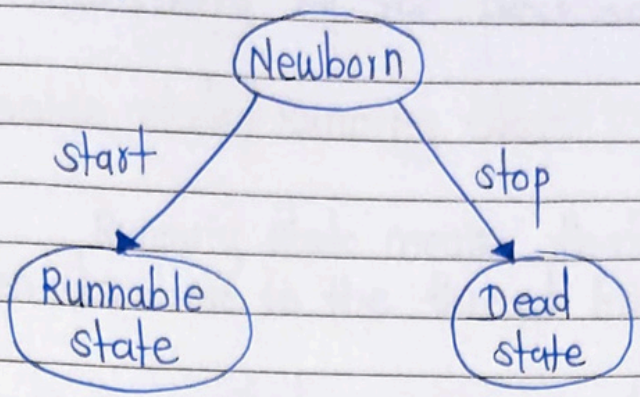
The thread can be in one state from there five states:

Newborn state:

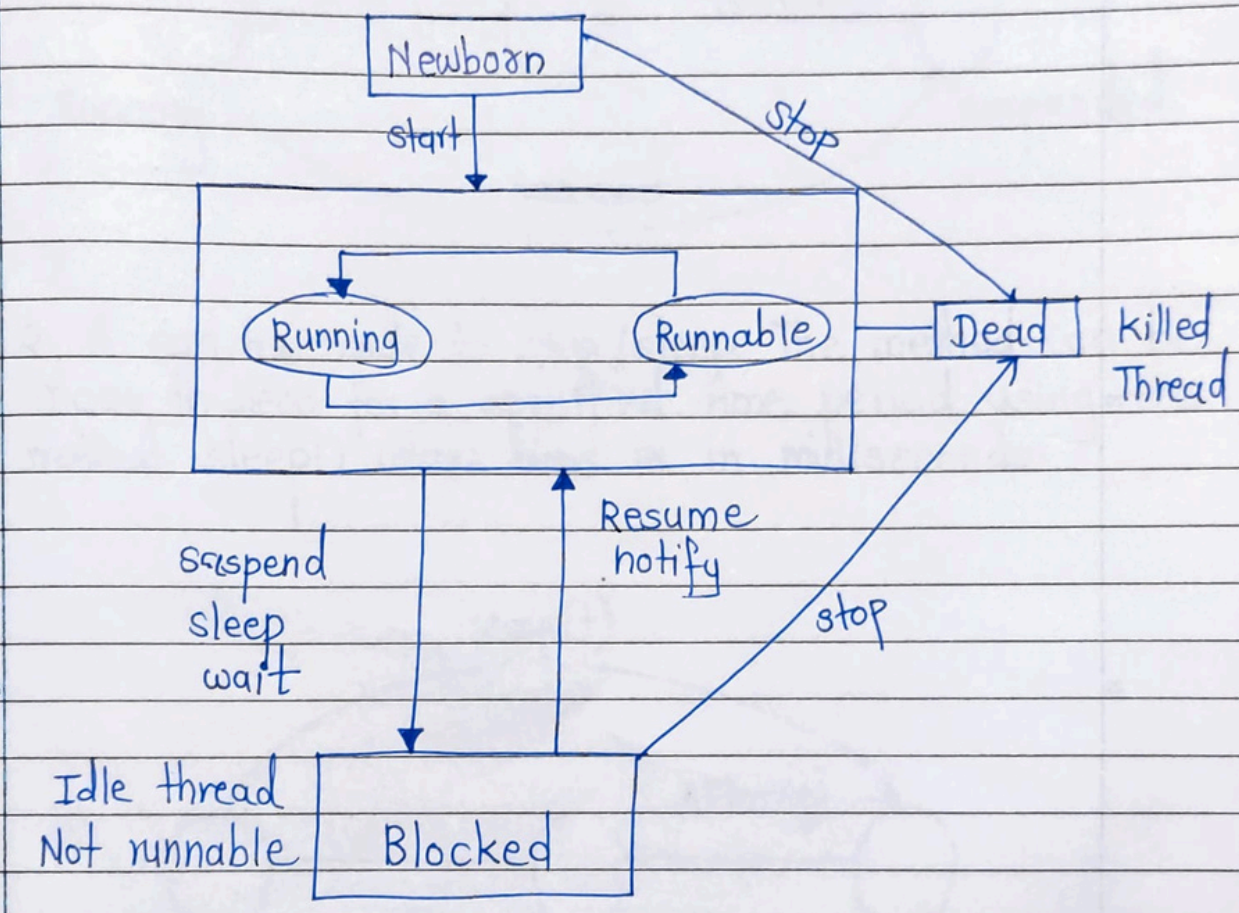
When a thread object is created, the thread is born and called as thread is in Newborn state.
When the thread is in newborn states if it can be either scheduled for running using start() method or kill it

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using stop methods.



• Five states of thread •



Runnable state:

The runnable state of a thread means that

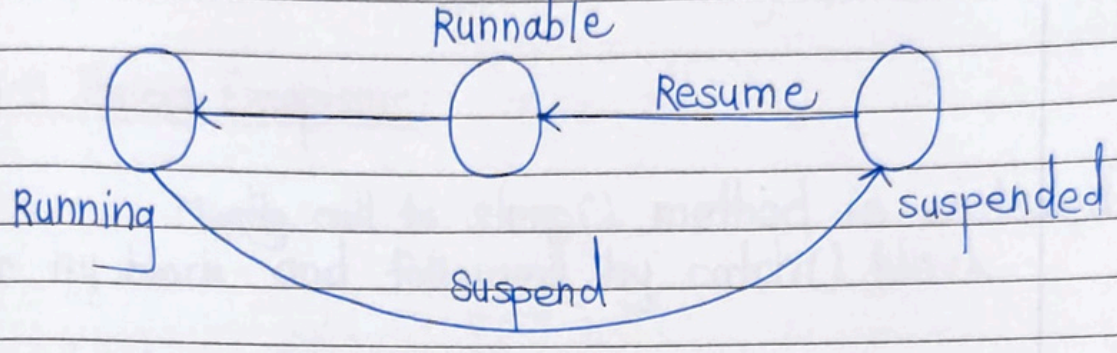
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the thread is ready for execution and is waiting for the availability of the processor.

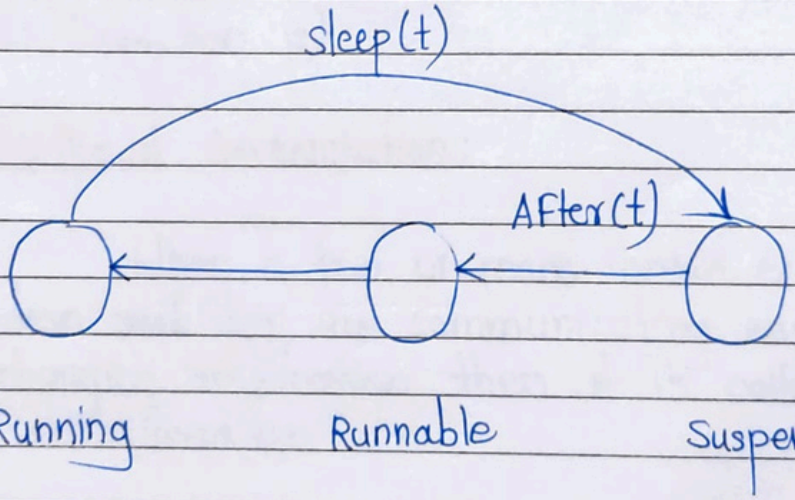
Runnable state / Running state:

Running state means that the processor has given its time to the thread for its execution.

1. It is suspended using suspended method.



2. It can be made to stop / sleep. The method can be made to sleep for a specified time period using the method sleep() where time is in milliseconds.



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Blocked state:

It is prevented from entering into the runnable state and subsequently the running state then it is said to be blocked state.

Dead state:

As every thread has a life cycle, a running threads ends its safe when it has completed executing its run() method.

14.3 Thread Exceptions:

Mostly call to sleep() method is enclosed in try block and followed by catch() block.

1. Catch (ThreadDeath e)

2. Catch (InterruptedException e)

3. Catch (IllegalArgumentException e)

4. Catch (Exception e)

14.4 InterThread Communication:

When a two or more thread exists in an application and they are communicating each other by exchanging information then it is called as inter thread communication.

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Following three methods are used:

1) wait()

2) notify()

3) notifyAll()

14.5 Deadlock

When two threads have a circular dependency on a pair of synchronized objects, then there is a deadlock.

When two or more threads are waiting to gain control of resource.

Due to some reasons, the condition on which the waiting threads rely on to gain control does not happen.

The result is called as Deadlock.

The code below illustrates deadlock

```
Thread A
    synchronized Method2() {
            .... synchronized Method1(){
        }
    }
    Thread B
    {
        synchronized Method1(){
            synchronized Method2() {
}}
```

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14.6 Errors and Exceptions :

Errors are wrongs or mistakes that can make a program go wrong.
Errors may be logical or may be typing mistakes.

An exception is an abnormal condition that arises in a code sequence at run time.

Types of Errors :

1) Compile time error

2) Run time error

Compile time errors

Java compilers detect all syntax error and display it. So these errors are known as compile time error

The common compile time errors :

1) missing semicolon

2) missing brackets in classes and methods

3) Misplelling of identifiers and keywords.

4) missing double quotes in strings

5) Bad reference to objects.

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Runtime error

After successfully compiling a program, class file is created but may not run properly like programs may produce wrong results due to wrong logic or may terminate due to errors such as stack overflow.

Exceptions:

An exception is a condition that is caused by a run-time error in a program.

Java keywords to manage exception:

1) try
2) catch
3) throw
4) throws
5) Finally

Exception handling Mechanism

The try keyword is used for the block of codes that causes an error condition and throws an exception. The catch block is added immediately after the key block.

```
try
{
   statement;
}
```

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```
catch (Exception e)
{
   statement;
}
   :
finally{

}
```

19.7 Builtin Exception

The standard java.lang. package defined several built in exceptions.
There are two types:

1) Unchecked Runtime Exceptions
2) checked Runtime Exceptions

Following list is Unchecked Runtime exceptions:

Exception	Description
1) Arithmetic exception	Divide by zero
2) ArrayIndexOutofBounds Exception	Array index out of Bounds
3) class cast Exception	Invalid cast
4) Illegal Argument Exception	Illiegal argument invoked.

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Following are checked Runtime Exception

Exception	Description
1) classNotFound Exception	class not found
2) IllegalAccessExeception	Access to class denied
3) InstatiationException	Attempt to create an object of an abstract class
4) NosuchFieldException	A requested field does not exist.

14.8 Throwing our own exceptions

Our own exception can be thrown by using the keyword throw as Follow:

```
throw new Throwable_subclass;
```

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